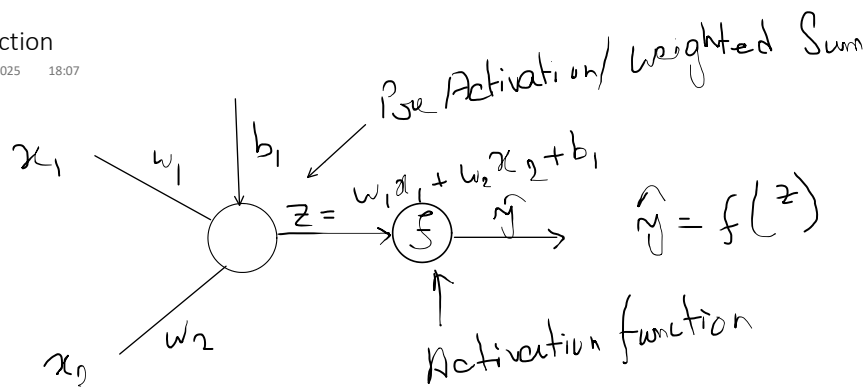
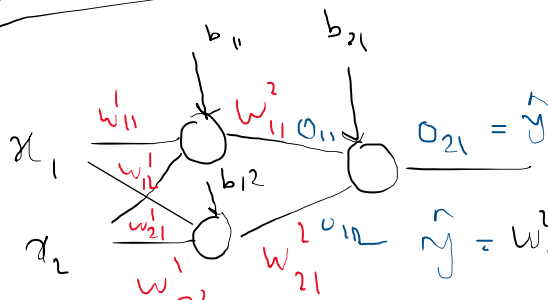




Why?

1. to capture non linearity without Act. Func.



$$\hat{y} = w_{11}^2 (w_{11}^1 x_1 + w_{21}^1 x_2 + b_{11}) + w_{21}^2 (w_{12}^1 x_1 + w_{22}^1 x_2 + b_{12}) + b_{21}$$

$$\hat{y} = w_{11}^2 w_{11}^1 x_1 + w_{11}^2 w_{21}^1 x_2 + w_{11}^2 b_{11} + w_{21}^2 w_{12}^1 x_1 + w_{21}^2 w_{22}^1 x_2 + w_{21}^2 b_{12} + b_{21}$$

Linear * without Activation function, non linearity cannot be captured *

Characteristics:

1. Non linear function

... linear ... → Desirable

1. Non linear → Desirable
2. Differentiable → Desirable
↳ Not mandatory (eg. ReLU)
3. Computationally inexpensive.
4. Zero centered → Desirable
↳ Not mandatory (eg. Sigmoid)
 $\text{mean}(\text{activation function of an layer}) = 0$
5. Non Saturating → Desirable
↳ Not mandatory (eg. tanh, Sigmoid)

Saturating for any large i/p
O/p will not be squished

$\left. \begin{array}{l} \text{tanh} \rightarrow [-1, +1] \\ \text{sigmoid} \rightarrow (0, 1) \end{array} \right\} \text{Vanishing Gradient Problem.}$

④ Farrow Activation function

Sigmoid | $\sigma(z) = \frac{1}{1 + e^{-z}}$

$$z = e^{-z}$$

Derivative

$$\sigma'(z) = \sigma(z) [1 - \sigma(z)]$$

$$\tanh'(z) = 1 - \tanh^2(z)$$

$$\tanh \mid \tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

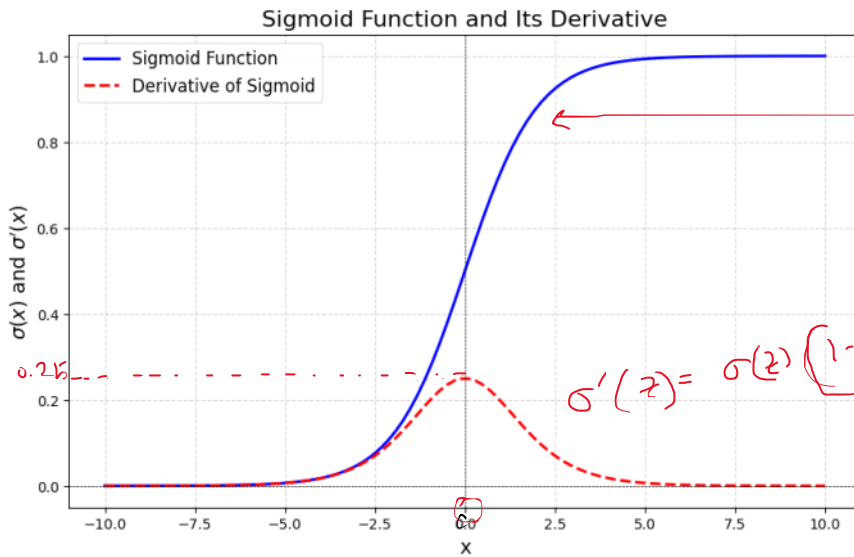
$$\text{relu} \mid \text{Relu} = \max(0, z)$$

$$\tanh'(z) = 1 - \tanh^2(z)$$

$$\text{relu}'(z) = \begin{cases} 0 & z \leq 0 \\ 1 & z > 0 \end{cases}$$

Sigmoid Activation Function

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Advantages:-

1. $(0, 1)$ ← treat as Probability

O/P layer ← Binary Classification

2. Non Linear

3. Differentiable

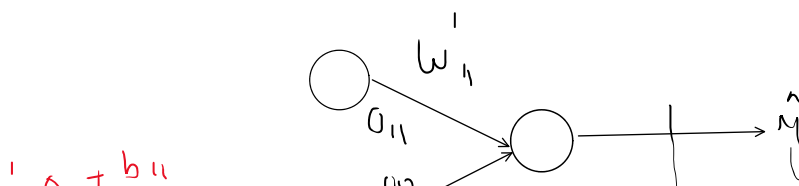
Disadvantages:-

1. Saturating function:- $\sigma(-\infty, +\infty) = (0, 1)$

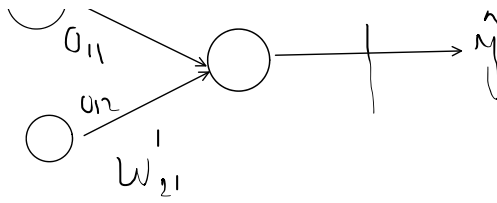
* Not used in Hidden layers.

↑
Vanishing Gradient

2. Non zero centered



$$z = w'_{11}O_{11} + w'_{21}O_{12} + b_{11}$$



$$\frac{dz}{dw'_{11}} = O_{11}$$

$$\frac{dz}{dw'_{21}} = O_{12}$$

$$O_{11} = \sigma(\text{Some})$$

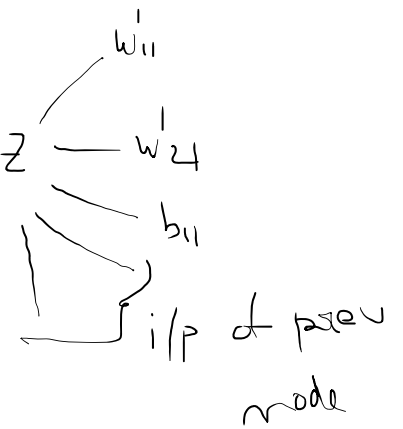
$$O_{12} = \sigma(\text{Some})$$

+/- ✓

$$\frac{dL}{dw'_{11}}$$

$$= \frac{dL}{dy} \cdot \frac{dy}{da} \cdot \frac{da}{dz}$$

$$\frac{dz}{dw'_{11}}$$



$$\frac{dL}{dw'_{21}}$$

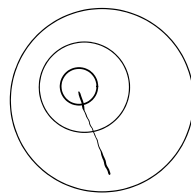
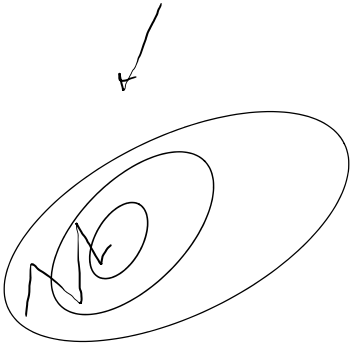
$$= \frac{dL}{dy} \cdot \frac{dy}{da} \cdot \frac{da}{dz}$$

$$\frac{dz}{dw'_{21}}$$

↑
+ve as it's $\sigma'(\text{Some})$

$W \rightarrow \begin{matrix} \uparrow \\ \downarrow \end{matrix} \left\{ \text{At once} \right\}$

$W \rightarrow \begin{matrix} \uparrow \\ \downarrow \end{matrix} \left\{ \text{Simultaneously} \right\}$



$$w'_i = w_i - \eta \frac{\partial L}{\partial w_i} \Rightarrow \underline{w'_i - w_i} = -\eta \frac{\partial L}{\partial w_i}$$

$$\Rightarrow \boxed{\Delta w_i = -\eta \frac{\partial L}{\partial w_i}}$$

$$\frac{\partial L}{\partial w_i} = \frac{\partial L}{dy} \cdot \frac{dy}{dz} \cdot \left(\frac{dz}{dw_i} \right)$$

↑
z = i/p or activation from prev layer.

$$\frac{\partial \mathcal{L}}{\partial w_i} = \frac{\partial \hat{y}}{\partial z} \left(\frac{\partial \mathcal{L}}{\partial z} \right)$$

Backpropagation Error $\rightarrow \delta$

prev layer $\rightarrow x_i$

$$\Delta w_i^o = -\eta \delta x_i^o$$

for non zero centered activation,

$$x_i^o > 0$$

$$\delta > 0 \Rightarrow \Delta w_i^o = (-) \quad (-ve)$$

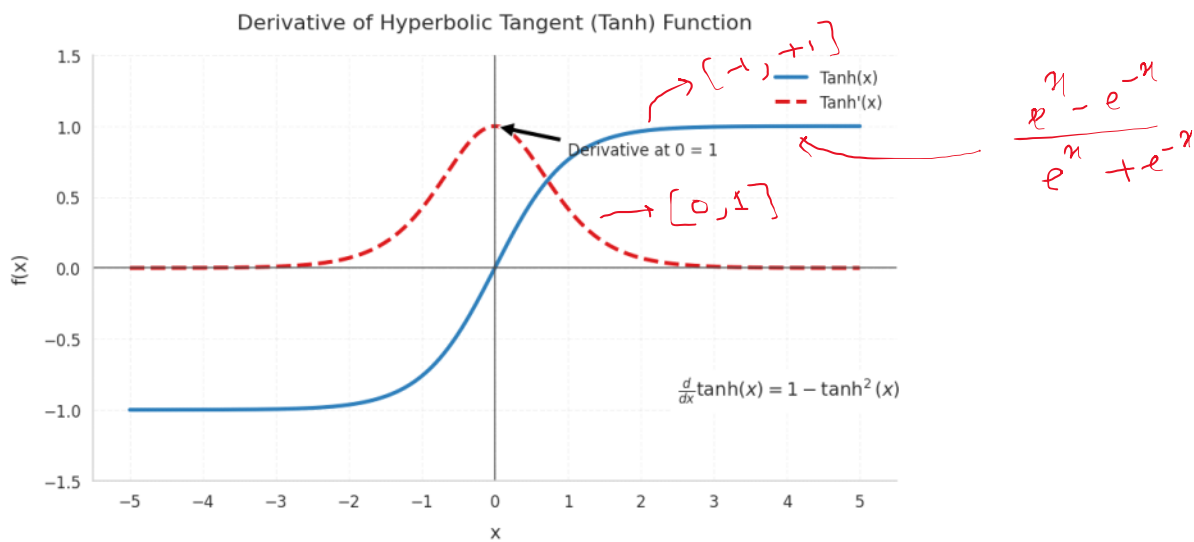
$$\delta < 0 \Rightarrow \Delta w_i^o = (+ve)$$

$\Delta w_i^o \rightarrow +ve$
 $\Delta w_i^o \rightarrow -ve$ } for same layer for all weights

3. Computationally expensive

Tanh Activation Function

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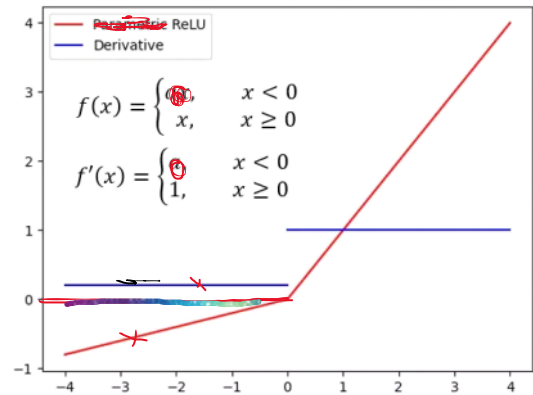
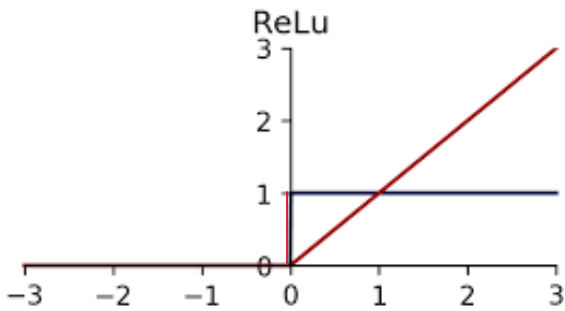
Advantage:-

1. Not linear
2. Differentiable
3. Zero centered (Smooth training)

Disadvantage:-

1. Saturating function (Vanishing Gradient Descent)
2. Computationally expensive

Rectified Linear Unit



$$f(x) = \max(0, x)$$

Advantages:-

1. Non Linear

$$f(x) = \max(0, x+1) - \max(0, x-1)$$

$f(x)$ is Linear

1. Additive Property

2. Homogeneity

$$f(x_1 + x_2) = f(x_1) + f(x_2)$$

$$f(ax) = af(x)$$

for relu

$$x_1 = -1 \quad a = 2$$

(1)

$$\text{Relu}(x_1 + x_2) = \text{Relu}(-1 + 2) = \text{Relu}(1) = 1$$

$$\text{Relu}(x_1) + \text{Relu}(x_2) = \text{Relu}(-1) + \text{Relu}(2) = 0 + 2 = 2$$

(11)

$$x = -2 \quad a = 3$$

$$\text{Relu}(ax) = \text{Relu}(3 \cdot -2) = \text{Relu}(-6) = 0$$

$$a \cdot \text{Relu}(x) = 3 \cdot \text{Relu}(-2) = 0$$

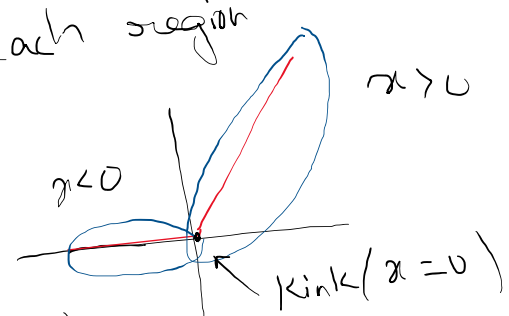
$$\alpha \text{Relu}(x) = \text{Relu}(-x) = 0$$

$$x=1 \quad \alpha=-1$$

$$\text{Relu}(\alpha x) = 0 \neq$$

$$\alpha \text{Relu}(x) = -1$$

* Relu is linear in each region
not globally



$$f(x) = \max(0, x+1) - \max(0, x-1)$$

$$\left. \begin{array}{l} = 0 \text{ when } x \leq -1 \\ = x+1 \text{ when } x > -1 \end{array} \right\} \begin{array}{l} = 0 \text{ when } x \leq 1 \\ = x-1 \text{ when } x > 1 \end{array}$$

Case I

$$x \leq -1$$

$$\max(0, x+1) = 0$$

$$\max(0, x-1) = 0$$

$$f(x) \stackrel{(-)}{=} 0$$

Case II

$$-1 < x \leq 1$$

$$\max(0, x+1) = x+1$$

$$\max(0, x-1) = 0$$

$$f(x) = x+1$$

Case III

$$x > 1$$

$$\max(0, x+1) = x+1$$

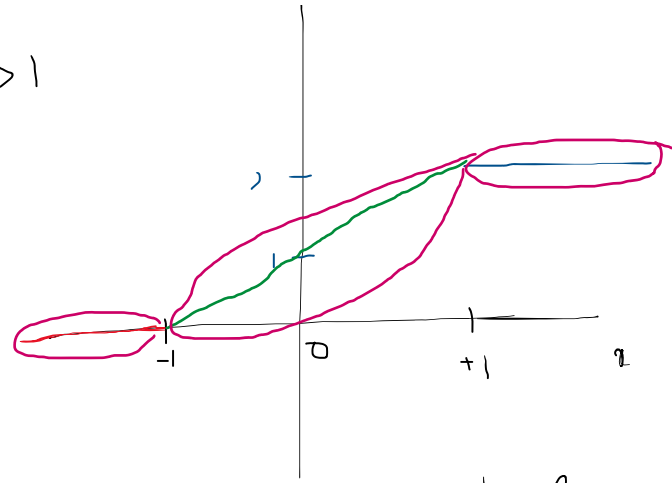
$$\max(0, x-1) = x-1$$

$$f(x) = 2$$

$$f(x) = \begin{cases} 0 & x \leq -1 \\ x+1 & -1 < x \leq 1 \\ 2 & x > 1 \end{cases}$$

$$h(x) = 2f(x+1) - 5f(x-1)$$

$x > 1$



Advantage:-

1. Non linear
2. Non Saturated in the +ve Region
3. Computationally inexpensive
4. Fast Convergence

Disadvantage

1. Not differentiable at $x=0$
2. Not ^{Batch Normalization} zero centered

3. Dying ReLU

Use
diff Variant
of ReLU

Dying Relu

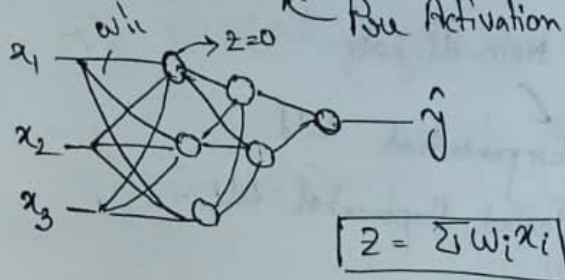
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19:11

Dying Relu :-

$$\text{Relu}(z) = \max(0, z)$$

Pre Activation / Weighted Sum.



$$z = \sum w_i x_i$$

if $z \leq 0$,

then $\text{Relu}(z) = 0$

$$\boxed{0_{11} = 0}$$

* if a node is dead once; then it is dead for ever.

$$z = f(w, b)$$

if 50% nodes are dead the the efficiency reduced to 50%.

Soln:-

1. Use small learning rate.

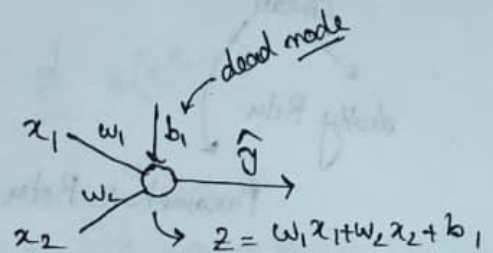
2. Use weight initialization techniques.

3. Use variants of relu

2. bias is very negative.

$$\boxed{z = w_1 x_1 + w_2 x_2 + b_1}$$

B. w is also very negative



Suppose $\boxed{z \leq 0}$

$$\text{Relu}(z) = 0$$

$$\frac{dL}{dw_1} = \frac{dL}{dy} \cdot \frac{dy}{dz} \cdot \frac{dz}{dw_1}$$

$$= 0$$

$$w'_1 = w_1 - \eta \frac{dL}{dw_1}$$

$$\Rightarrow w'_1 = w_1 - \eta \times 0 \Rightarrow \boxed{w'_1 = w_1}$$

no training

* when z becomes ≤ 0 if learning rate is too high.

$$w'_1 = w_1 - \eta \frac{dL}{dw_1}$$

$$w'_2 = w_2 - \eta \frac{dL}{dw_2}$$

$$b'_1 = b_1 - \eta \frac{dL}{db_1}$$

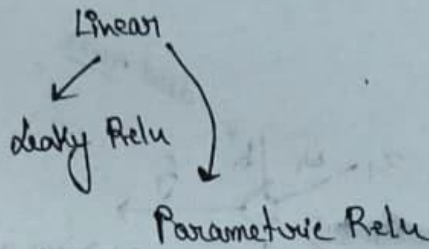
⇓

-ve $\leftarrow$ small - large
 $z = f(w, b)$

Variants of Relu

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Variant of Relu:-



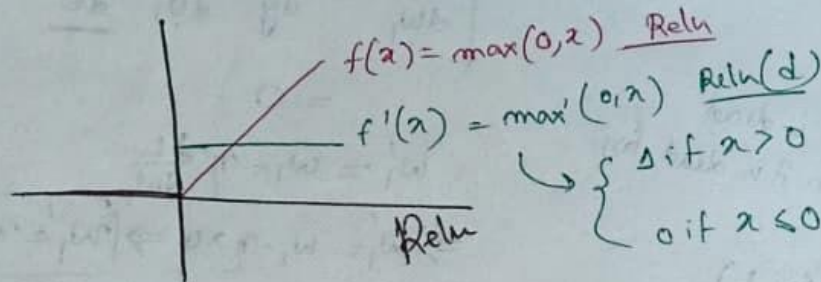
Non linear

Exponential LU

Scaled Exponential LU

Leaky Relu:-

$$\text{Relu} \rightarrow f(x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$



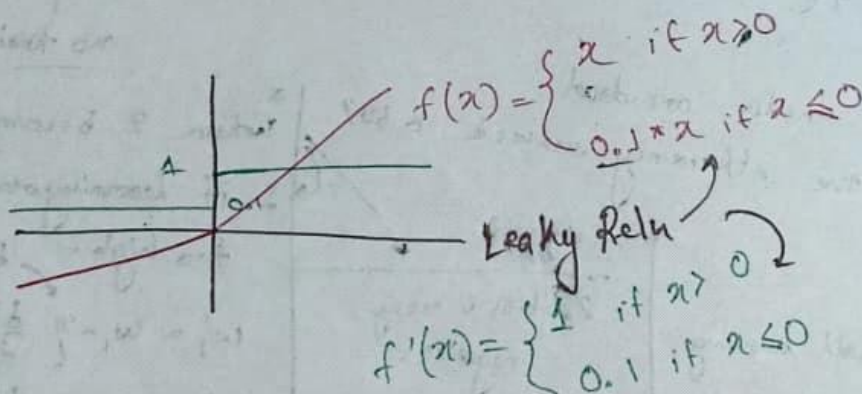
advantage

i) No dying Relu Problem

ii) No Saturation

iii) Close to Zero Centred

iv) less Computation



Parametric Relu:-

$$p\text{Relu} = f(x) = \begin{cases} x & \text{if } x > 0 \\ a * x & \text{if } x \leq 0 \end{cases}$$

$$f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ a & \text{if } x \leq 0 \end{cases}$$

Trainable Parameters.

Adv

i) Adv of LReLU

ii) flexibility.

① Exponential Linear Unit:-

$$ELU(x) = \begin{cases} x & \text{if } x > 0 \\ a(e^x - 1) & \text{if } x \leq 0 \end{cases} \quad \text{hyper param.}$$

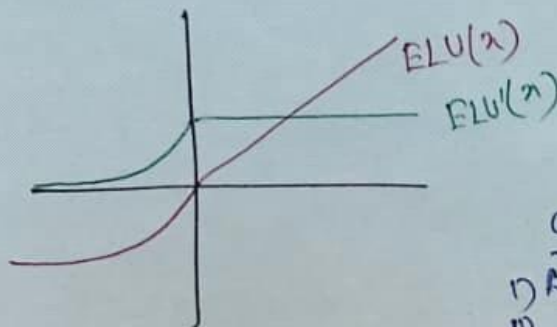
$$ELU'(x) = \begin{cases} 1 & \text{if } x > 0 \\ ELU(x) + a & \text{if } x \leq 0 \end{cases} \quad \left| \quad \frac{d}{dx} a(e^x - 1) \right.$$

$$= ae^x + 0$$

$$= ae^x - a + a$$

$$= a(e^x - 1) + a$$

$$= ELU(x) + a$$



advantage:

- i) Adv. of prev. variants
- ii) No linear in negative reg.

disadvantage:-

- i) Computationally expensive
- iii) Better generalisation

② Sealed ~~Linear~~ Exponential Linear Unit:-

$$SELU(x) = \lambda \begin{cases} x & \text{if } x > 0 \\ a(e^x - 1) & \text{if } x \leq 0 \end{cases}$$

$$\hookrightarrow \begin{cases} \lambda x & \text{if } x > 0 \\ \lambda a(e^x - 1) & \text{if } x \leq 0 \end{cases}$$

$$SELU'(x) = \begin{cases} \lambda & \text{if } x > 0 \\ \lambda a e^x & \text{if } x \leq 0 \end{cases} = \lambda \begin{cases} 1 & \text{if } x > 0 \\ a e^x & \text{if } x \leq 0 \end{cases}$$

a } const
 λ }

③ a_1

④ a_2

⑤ a_3

⑥ a_m

$$\text{mean}(a_1, a_2, a_3, \dots, a_m) = 0$$

disadv.

- i) disadv. of ELU.

5